Section

EXERCISE 1: Robot Kinematics

Consider the robot in the picture and the associated equations.

$$v = \frac{V_r + V_l}{2}$$

$$\omega = \frac{V_r - V_l}{b}$$

1

What are the formulas about?
(-/1 punti)

those formulas refer to differential drives kinematics

2

What does b refer to in the formulas? What happens if you measure it longer than it actually is in the real robot?
(-/2 punti)

b refers to the base line which is the distance between the 2 wheels. if you measure it longer than it is you will have that the robot steers more than you want because you will get an higher wz than expected.

3

What happens if the robot has the right tyre more inflated than the left one?
(-/1 punti)

if the right tire is more inflated than the left one you will obtain that the radius of the right tire is bigger than the left one. so, if you are steering left, you will steer more than expected because Vr depends on the radius of the right wheel and if you have an higher Vr you will get an higher wz. if you are steering right you will steer less than expected.

4

What is the ICC? Where the ICC of this robot can be? Are there special locations?
(-/2 punti)

the ICC is the instantanous center of curvature (also known as instantaneous ceter of rotation ICR). It is the point where all the axes of the wheels intersect while the robot is steering. in this robot the ICC is aligned with the two wheels at a distance of R from the point in the middle of the 2 wheels where R is the curvature radius. there are two special locations: when the robot is rotating in place the ICC is in the middle of the two wheels (R=0), when the robot is going straight the ICC tends to infinite.

Section

EXERCISE 2: Localization

In the following we assume we want to localize a robot dog in an arena surrounded by 6 colored posts in known positions. One of the possible implementation of the Bayes filter for localization is the Kalman Filter (KF) or its non linear version known as the Extended Kalman Filter (EKF).

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5

Write the sensor measurement model in this case and discuss it in details. Be practical in the answer as you should be ready to implement it yourself.
(-/2 punti)

The senosr model represnt the probability to get a certain value after a meausrement in a certain position. if you get more than one measurement in one scan of the envoironment you can consider them independet. so the P(z|x)=pi of i 1->n of P(zi|x); the sensors can get measurments that can be influnced by obstacles, random measurement, noise and out of range values. so you can represent a measurement like a gaussian distribution around the expected value adding uniform distribution for noise and out of range values and adding an hyperbolic function untill the expected value to represent the obstacles.

6

Do we need an Extended Kalman filter or a Kalman filter is enough? Why?
(-/1 punti)

we need an EKF because KF deals only with linear models but we have a non linear model for the sensor measurements. EKF can deals with non linear models.

7

Build the Extended Kalman Filter (EKF) selecting the correct steps (NOTE: Multiple choices apply)
(-/3 punti)

- [x] mu' = g(mu,u)
- [ ] mu' = G * mu
- [ ] mu' = G * mu + B * u
- [x] Sigma' = G^T * Sigma * G + R
- [x] K = Sigma * H^T * (H * Sigma' * H^T +Q)^-1
- [ ] K = Sigma' * C^T * (C * Sigma' * C^T +Q)^-1
- [x] mu = mu' + K(z - h(mu'))
- [ ] mu = mu' + K(z - H * mu')
- [x] Sigma = (I - K * H) * Sigma'
- [ ] Sigma = (I - K * C) * Sigma'

8

What are H and G in the previous question?
(-/1 punti)

H and G are the hessians of the function xt = g(xt-1,u) and zt = h(xt). they represent the derivative of the functions wrt their variables and are used to obtain a first order tayolr approximation of the system

Section

EXERCISE 3: Simultaneous Localization and Mapping

Exercize 4: Robot Operating System

Let's assume we want to build an exam proctoring system based on several cameras monitoring students doing their robotics exam at home.

11

Could we use ROS to implement it? How? ... I just need the rationale behind your answer here.
(-/2 punti)

yes we can. to do this we can consider each student's pc as a publisher node that retrive data from the webcam and publish them on a topic as a custom messages with student's personal info. then we can consider professor's pc as a subscriver node that retrive data from the topic where the students are publishing.

12

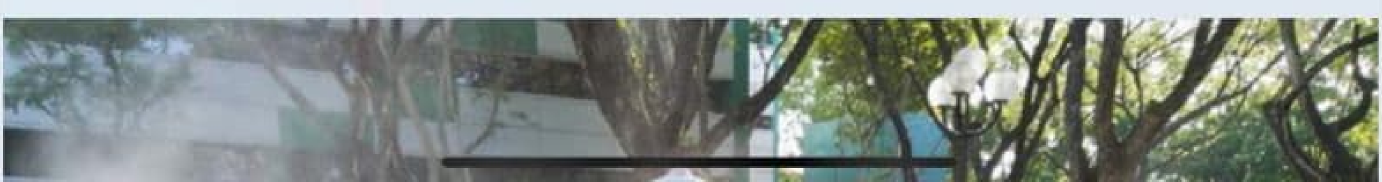
What would you use services for in your system? Make an example!
(-/2 punti)

we can use servers for example to analyze suspicious behaviours. once the professors node has recived all the data from a student node it can send a request to a server to analayze the images and discover for example if the student looked out of the screen several times or something like that. once the server has computed the evaluation it sends back the result to the professor's node (which is the client).

Section

EXERCIZE 5: Robot Motion Control

Let's consider a sanitizing wheeled robot robot which each night has to traverse a university campus spraying around disinfectant to sanitize the areas before the students can access them.



Section

EXERCIZE 5: Robot Motion Control

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13

What are the characteristics of your robot in terms of kinematics and payload? Why?
(-/2 punti)

the robot in the picture has an ackerman steering kinematics, which is the most diffused kinematics in the world. In terms of payload the robot will be very heavy because of the liquid. the center of gravity will be probably near the tank where the liquid is stored. because of the weight we need a really solid structure to support the liquid and the chassis of an ackerman steering can support it really well.

14

14

Which of the obstacle avoidance algorithms would you chose among VHF, VHF+, and DWA for your robot? Why?
(-/2 punti)

we need to consider also the dynamics of the robot becuse, due to the liquid, when it's at full charge it's not neglectable. the best way to consider the robot's dynamics is to use the DWA algorithm because it takes into account the acceleration with the dynamic window. considering acceleration we can avoid robot crashes dued to inertia of the liquid.

15

Let assume we want to control a drone in an indoor environment. How would you implement its obstacle avoidance system? Why?
(-/2 punti)

to implement it's avoidance system we need sensors pointing up and sensors pointing forward. we can simplify the

16

to implement it's avoidance system we need sensors pointing up and sensors pointing forward. we can simplify the

16

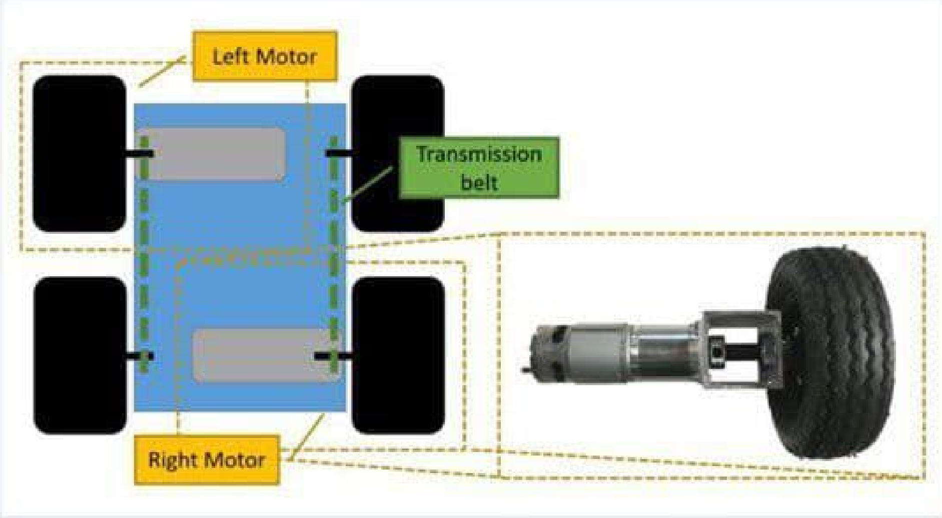
Is a drone a good idea to solve the problem posed in this exercise? Why?
(-/1 punti)

I think that a drone is not a good solution to solve the problem because of the weight of the liquid. if we want to carry a big tank we need a really big drone but big drones aren't advisable for indoor applications. otherwise you can carry a small tank but you have to refill it continuously. with a wheeled robot you haven't this problem: the robot can carry a big tank without problems.

Section

EXERCISE 1: Robot Kinematics

Consider the top view robot in the picture with two motors like in the figure. The front direction is the one on the top part of the figure.



1

What kind of kinematics has the robot in the picture? Why?
(-/1 punti)

Since the two wheels on each side are coupled by the transmission belt, we can affirm the robot in the picture has a differential drive kinematics.

2

Report the forward kinematics equations for the robot in the picture and discuss their parameters.
(-/2 punti)

V = (Vr + Vl)/2.
Omega = (Vr - Vl)/L.
R = [L*(Vr + Vl)] / [2*(Vr - Vl)].
Vr and Vl are respectively the right and the left wheel velocity. L is the distance between the wheel centers. R is the curvature radius. In this case we'll refeer to the actuated wheels velocity.

3

Design a calibration procedure to estimate the parameters of the forward kinematics of the robot in the picture
(-/1 punti)

The paramater to know, and which could affect the model later if uncertain is L. In

3

Design a calibration procedure to estimate the parameters of the forward kinematics of the robot in the picture
(-/1 punti)

The paramater to know, and which could affect the model later if uncertain is L. In order to estimate it right, i could make the robot perform different curves, recording all the velocities, and then obtain L as the value that maximies the probability of having the recorded values.

4

Provide two example applications this robot is suited for? Why?
(-/2 punti)

This kind of robots is suited for the motion in narrow and/or crowded places. Two examples of application are: the house cleaning (Roomba), an autonomous wheel

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This kind of robots is suited for the motion in narrow and/or crowded places. Two examples of application are: the house cleaning (Roomba), an autonomous wheel chair.

Section

EXERCISE 2: Localization

In the following we assume we want to localize a robot dog in an arena surrounded by 6 colored posts in known positions and with known color codes. Different solutions exists with pros and cons implementing the Bayes filter for localization. For each of them, and with reference to this specific scenario, provide a brief description of the functioning, a brief description of the basic elements, and a brief description of the limits.

5

How would you implement it with a Discrete filter?
(-/2 punti)

Let's start with assumptions for the whole exercise: the arena is small, the number of landmark is given and it is low, also the motion model is probaby non linear (so the normal Kalman Filter would not be enough); finally, on the base of the color code of the landmarks, i would use computer vision in the sensoring part. I would implement discrete filter in such way: first i would divide in cells the arena, then, after performing movement, i would take note of the observed landmark; iterating the procedure i should obtain the robot position. Since the arena is small and the landmarks are well codified, discrete filter should already give a good result.

6

How would you implement it with an Extended Kalman Filter?
(-/2 punti)

In this case i would first need the initial state of the robot, then i would model the uncertainity on the state of the robot with a Gaussian with average mu and covariance sigma; after have computed the jacobian matrix from the non linear model of the robot, i would iterate alternating a first a prediction on mu and sigma based on the motion model, and then perform a correction basing on the Kalman gain, to weight the confidence with the sensor model. The EKF it's limited usually by the the Gaussian Distribution assumption and by unimodality, but these weak points should be balanced for the same reason the discrete filter should give a good result.

7

How would you implement it with a

7

How would you implement it with a Particle Filter?
(-/2 punti)

With particle filter i can go beyond the weak points of the EKF, thanks to the filter being multimodal. One advantage is that i could implement solution also for the kidnapped robot problem. I would implement particle filter in such way: first i would consider a density of probability for the robot to be everywhere in the arena, then, alternating between the landmarks observed and the motion performed, i would update the density function accordingly, to finally converge on the robot position (the particle would finally gather around the robot correct position).

Section

EXERCISE 3: Simultaneous Localization and

What is the motion model and what it is used for.
(-/2 punti)

The motion model is based on the kinematics of the robot and describes how the robot performs motion based on its state and the input action. It is used to make a first estimate on the robot position on the base of the previous state, that will be corrected in a second step.

9

What is the measurement model and what it is used for.
(-/2 punti)

The measurement model describes the bond between the measurement the robot gives back and the pose from which that measurement has been done. It is used in order to perform finally a correction on the robot state estimation previously estimated by the means of the motion model.

10

What is TF? What is it used for?
(-/2 punti)

TF is a package used in order to keep track of multiple coordinate frame over time. It describes the relationship between coordinate frames with a tree structure and gives the user the possibility to transform points and vectors from one frame to another coordinate one at any time.

11

When should you use services and actions instead of the standard topics mechanism?
(-/1 punti)

I should do it when what i want to be performed can be seen as a subroutine with respect to the rest of the network.

12

Omnidirectional Kinematics
(-/2 punti)

DWA, since omnidirectional kinematics works better with curves and DWA considers only circular trajectories.

13

Skid Steering Kinematics
(-/2 punti)

VHF, since i expect to use skid steering kinematics for linear or almost linear motion.

14

Differential Drive kinematics
(-/2 punti)

14

Differential Drive kinematics
(-/2 punti)

DWA with roll-out, in order to consider both circular and not circular trajectories, since differential drive works fine with both kind of trajectories.

15

Ackerman Kinematics
(-/2 punti)

VHF+: since with ackerman kinematics wheels can perform limited turning angles, i would implement a motion control algorithm that works better with curves to respect with VHF, but letting the other algorithms to other kinematics more suited for circular trajectories.